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PT. TRISTAR MARINE

OPERATING PROCEDURE

TM-WP-016

HULL SURVEY

THIS AND ALL STANDARD PRACTICE INSTRUCTIONS/OPERATING PROCEDURES CONTAINED IN THE COMPANY QUALITY ASSURANCE SYSTEM ARE CONTROLLED BY THE QUALITY ASSURANCE MANAGER AND MAY NOT BE AMENDED, REVISED OR IN ANY WAY WITHOUT HIS AUTHORITY.

The signatures below indicate Preparation, Review and Approval of all the numbered pages in this Procedure up to and including the date of approval. This Procedure is invalid without page 1

Date	Rev.	Written	Reviewed	Approved	Signature	Revision Detail
22.5.14	0	Caesar Aqsa	Isran	Isran		Original Issue

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8.0 THICKNESS MEASUREMENT of HULL SCANTLING

1.0 SCOPE

This procedure provides guidance to PT. Tristar Marine personnel conducting hull survey whenever compliance with IACS is required.

2.0 PERSONNEL

Personnel conducting hull survey shall be qualified

3.0 EQUIPMENT

For hull surveying purposes a few personal standard parts are necessary for the surveyor, whereas for special inspections additional tools or checking devices may be required.

Surveyors must be familiar with the equipment and tools used in hull survey procedures. Ship equipment can be used, and in addition, a shipyard may provide necessary devices. Before using measuring devices, calibrations and zero adjustments should be carried out, if necessary.

3.1. Standard Equipment Personal Property

- Suitable working clothes
- Safety shoes
- Safety helmet
- Ear protectors
- Gloves
- Lighting torch
- Test hammer
- Pocket measuring tape

3.2. Other Tools and Equipment Used in Survey

➢ Testing hammer:

Note that hammers made of steel must not be used in gas dangerous or gas suspected areas. Here the use of alloy hammers/tools is required to avoid that sparks are initiated by hammering or when tools slip out of hand and fall down.

➢ Measuring tools:

Dial gauges, welding gauges, and feelers may be required for measuring clearances, welding seam heights, indents or deflections of hatches etc.

- Portable oxygen and explosive content meters.

Portable oxygen and explosive content meters are necessary in areas where petroleum liquids or others are carried and where compartments with limited oxygen content shall be inspected. If ship's instrument is used, proper calibration of the instrument should be confirmed by a suitable testing method or by the latest calibration record issued by a recognized testing firm.

- Thickness measurement instruments:

Such instruments use ultrasonic testing methods and calibration of the instrument is usually built in, otherwise it must be gauged separately.

- Vibration measurement equipment

Vibration measurement equipment usually available through a shipyard or at major offices or testing institutes is to be used with reference to the relevant instructions and application procedures. There are numerous devices developed and suitable for hull vibration measurements and for recording vibrations exercised by running machinery.

- Camera:

Photographs can greatly aid proper reporting if a suitable camera is available. The use of flashlights, however, is limited in areas where explosion risk exists.

4.0 SAFETY

Before commencing any survey, the owner and all personnel involved in the survey must ensure that appropriate safety procedures are specified.

Safety standards often vary from owner to owner and vessel to vessel. While many owners maintain strict onboard safety procedures, survey personnel including Classification Society surveyors and subcontractors must still be aware of what constitutes minimum acceptable standards for tanker inspection to protect themselves from potentially hazardous working conditions.

5.0 SURVEY PROCEDURES: GENERAL

Hull survey methods, are means and procedures to detect failure and damage at an early stage to avoid premature breakdown where the details of structural components are within the close visual inspection range of the surveyor i.e. normally within reach of hand.

Hull survey methods are therefore not only comprehensive means of detecting deficiencies or monitoring structural condition, but also of defining schemes for inspection between the last overhaul and before the occurrence of failure.

Means of detection of defects and condition monitoring are inter alia:

- Visual inspections.
- Non-destructive testing (NDT).
- Examination of tightness, function test.
- Measurements of thickness, vibration.

Schemes of inspection are periodical survey requirements which by virtue of design and operational experience are envisaged to discover deficiencies completely and early enough before they may lead to breakdown.

6.0 VISUAL INSPECTION

A major part of hull surveying work is carried out using visual skills to perform the examinations and to arrive at an opinion on the state of a vessel's condition.

Such visual examinations can be carried out as:

- Over-all inspections, a general sighting of a vessel's hull condition, followed by
- Close-up examinations at.
  - i) Locations where discontinuities, ruptures or deformations have been found and
  - ii) Certain hull structures as stipulated by rules and/or requirements, for instance in way of cargo area of oil tanks.
- Examination of areas of suspected crack and corrosion concentration.

7.0 NON-DESTRUCTIVE TESTING METHODS

The detection of cracks by visual methods is rather limited. Additionally, internal welding seam imperfections or flaws in material parts cannot be discovered without suitable means of examination and instrumentation. To discover these suitable means of non-destructive testing (N.D.T.) are used, such as:

- Dye checks with liquid penetrants
- Magnetic particle checks
- Radiographic checks
- Ultrasonic measurements.

8.0 THICKNESS MEASUREMENT of HULL SCANTLINGS

In general, thickness measurements are made by ultrasonic thickness gauges.

If carried out professionally and in a representative way, measurements of the actual thickness of scantlings can generally reveal the actual overall condition of a vessel with respect to its structural strength.

The scope of the measurements required is determined by the rules of the classification society based on the type and age of a vessel under survey. The actual conditions of the structure, verified by visual observations, may request premature and/or additional measurements.

As a general rule, the smaller the thicknesses the more the extensive measurements have to be.

In areas of heavy corrosion testing is to be increased to show the extent of wear and to allow proper judgment if the area is to be renewed or otherwise repaired.

9.0 PREPARATIONS FOR SURVEY

9.1. CONDITIONS FOR SURVEY

9.1.1. The Owner is to provide the necessary facilities for a safe execution of the survey.

9.1.2. Tanks and spaces are to be safe for access, i.e. gas freed, ventilated and illuminated.

9.1.3. In preparation for survey and thickness measurements and to allow for a thorough examination, all spaces are to be cleaned including removal from surfaces of all loose accumulated corrosion scale. Spaces are to be sufficiently clean and free from water, scale, dirt, oil residues etc. to reveal corrosion, deformation, fractures, damages, or other structural deterioration. However, those areas of structure whose renewal has already been decided by the Owner need only be cleaned and descaled to the extent necessary to determine the limits of the areas to be renewed.

9.1.4. Sufficient illumination is to be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration.

9.1.5. Where soft or semi-hard coatings have been applied, safe access is to be provided for the surveyor to verify the effectiveness of the coating and to carry out an assessment of the conditions of internal structures which may include spot removal of the coating. When safe access cannot be provided, the soft or semi-hard coating is to be removed.

9.2. ACCESS TO STRUCTURES

9.2.1. For survey, means are to be provided to enable the surveyor to examine the hull structure in a safe and practical way.

9.2.2. For survey in cargo holds and water ballast tanks, one or more of the following means for access, acceptable to the Surveyor, is to be provided:

- permanent staging and passages through structures;
- temporary staging and passages through structures;

- lifts and movable platforms;
- boats or rafts;
- other equivalent means

10.0 SURVEY REPORTING PRINCIPLES

As a principle, the surveyor is to include the following content in his report for survey of hull structure and piping systems, as relevant for the survey.

10.1. GENERAL

10.1.1. A survey report is to be generated in the following cases:

- In connection with commencement, continuation and / or completion of periodical hull surveys, i.e. annual, intermediate and special surveys, as relevant
- When structural damages / defects have been found
- When repairs, renewals or modifications have been carried out
- When condition of class (recommendation) has been imposed or deleted

10.1.2. The purpose of reporting is to provide:

- Evidence that prescribed surveys have been carried out in accordance with applicable classification rules.
- Documentation of surveys carried out with findings, repairs carried out and condition of class (recommendation) imposed or deleted
- Survey records, including actions taken, which shall form an auditable documentary trail.
- Information for planning of future surveys.
- Information which may be used as input for maintenance of classification rules and instructions.

10.2. EXTENT OF SURVEY

10.2.1. Identification of compartments where an overall survey has been carried out.

10.2.2. Identification of locations, in each tank, where a close-up survey has been carried out, together with information of the means of access used

10.2.3. Identification of locations, in each tank, where thickness measurement has been carried out.

10.3. RESULT OF THE SURVEY

10.3.1. Type, extent and condition of protective coating in each tank, as relevant (rated GOOD, FAIR or POOR).

10.3.2. Structural condition of each compartment with information on the following, as relevant: Identification of findings, such as:

- Corrosion with description of location, type and extent

- Areas with substantial corrosion
- Cracks / fractures with description of location and extent
- Buckling with description of location and extent
- Indents with description of location and extent

Identification of compartments where no structural damages / defects are found

The report may be supplemented by sketches / photos.

11.0 ATTACHMENTS

- IACS Minimum Requirement to Close Up Surveys at Oil Tankers
- Hull Survey Report [FORM-WP-016-01]

IACS Minimum Requirement to Close Up Surveys at Oil Tankers

Z10.1

Z10.1
(cont'd)

TABLE I

Table of Minimum Requirements to Close-up Surveys at Special Survey of Oil Tankers, Ore/Oil Ships and etc.

Special Survey No.1 age ≤ 5	Special Survey No.2 5 < age ≤ 10	Special Survey No.3 10 < age ≤ 15	Special Survey No.4 and Subsequent age > 15
A) ONE WEB FRAME RING - in a ballast wing tank, if any, or a cargo wing tank used primarily for water ballast	A) ALL WEB FRAME RINGS - in a ballast wing tank, if any, or a cargo wing tank, used primarily for water ballast	A) ALL WEB FRAME RINGS - in all ballast tanks	As special survey No.3
B) ONE DECK TRANSVERSE - in a cargo oil tank	B) ONE DECK TRANSVERSE - in each of the remaining ballast tanks, if any	A) ALL WEB FRAME RINGS - in a cargo wing tank	Additional transverses included as deemed necessary by the Classification Society
D) ONE TRANVERSE BULKHEAD - in a ballast tank	B) ONE DECK TRANSVERSE - in a cargo wing tank	A) A minimum of 30% of all web frame rings in each remaining cargo wing tank (see Note 1)	
D) ONE TRANSVERSE BULKHEAD - in a cargo oil wing tank	B) ONE DECK TRANSVERSE - in two cargo centre tanks	C) ALL TRANSVERSE BULKHEADS - in all cargo and ballast tanks	
D) ONE TRANSVERSE BULKHEAD - in a cargo oil centre tank	C) BOTH TRANSVERSE BULKHEADS - in a wing ballast tank, if any, or a cargo wing tank used primarily for water ballast	E) A minimum of 30% of deck and bottom transverses including adjacent structural members in each cargo centre tank	
	D) ONE TRANSVERSE BULKHEAD - in each remaining ballast tank	F) As considered necessary by the surveyor	
	D) ONE TRANSVERSE BULKHEAD - in a cargo oil wing tank		
	D) ONE TRANSVERSE BULKHEAD - in two cargo centre tanks		

A) Complete transverse web frame ring including adjacent structural members
B) Deck transverse including adjacent deck structural members
C) Transverse bulkhead complete – including girder system and adjacent structural members
D) Transverse bulkhead lower part – including girder system and adjacent structural members
E) Deck and bottom transverse including adjacent structural members
F) Additional complete transverse web frame ring

See sketches in Sheet 15.

Note 1: The 30% is to be rounded up to the next whole integer.

Z10.1 Annex II

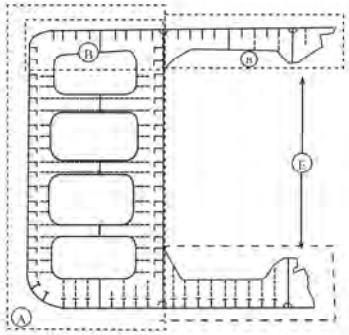
Sheet 15

Z10.1
(cont'd)

Thickness Measurement - Oil Tankers, Ore/Oil ships and etc.

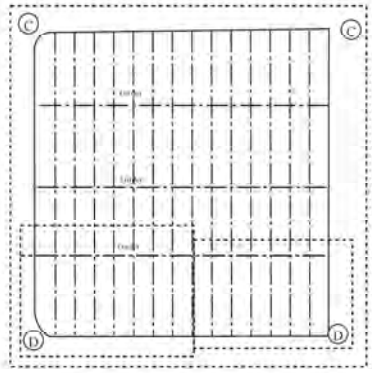
Close-up survey requirements

Oil Tanker
Typical transverse section close-up survey



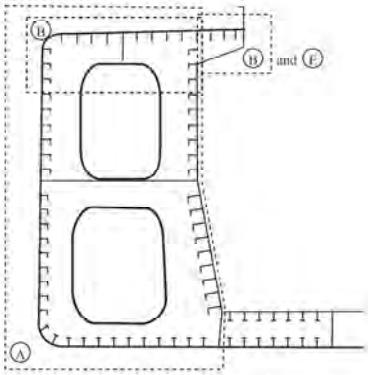
Thickness to be reported on
TM3-T and TM4-T as appropriate

Oil Tanker
Typical transverse bulkhead



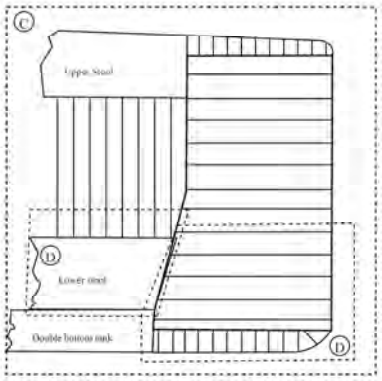
Thickness to be reported on TM5-T

Oil/Ore ship
Typical transverse section close-up survey



Thickness to be reported on
TM3-T and TM4-T as appropriate

Oil/Ore ship
Typical transverse bulkhead



Thickness to be reported on TM5-T

Close-up survey area

Recommendations for the extent and pattern of gaugings are indicated in Table IV of the IACS Unified Requirements

HULL SURVEY REPORT

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HULL SURVEY REPORT

Report No.: . . . . . .

GENERAL PARTICULARS	
Name of Ship: Nama Kapal	Ship Owner: Pemilik Kapal
Ship Type: Tipe Kapal	IMO Number: Nomor IMO
Port of registry: Pelabuhan terdaftar	National Flag: Bendera
Date of Build: Tahun Pembangunan	Builder: Gal. Pembangun
Classification of ship: Klasifikasi kapal	Class ID: ID Klasifikasi
Deadweight (MT): Bobot Mati (MT)	G.R.T (ITC 69): Tonnase Kotor

a) The survey reports and documents listed below have been reviewed by the undersigned and found to be satisfactory

b) A summary of the survey is attached herewith on sheet 2

c) The hull special survey has been completed in accordance with the Regulations on [date]

Prepared by:	Acknowledged by:	Approved by:
(. . . . .)	(. . . . .)	(. . . . .)

Form-WP-016-1

Naval Architect & Offshore Engineering, FSRU, FSO, FPSO Supervision, Ship Survey & Assessment, Ship Maintenance Management, Ship Re-drawing, Cargo Gear Load Test, Hull Thickness Measurement, Non-Destructive Test, Rope Access Services, Underwater Survey/ROV/salvage and Drone Survey.
Approved by: ABS, BKI, BV, DNV-GL, IRS, LR, KR, NK, RINA

TRISTAR MARINE

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EXECUTIVE HULL SUMMARY

A.	**Report Review**	-	Where and how survey was done
B.	**Close-up Survey**	-	Extent (Which tanks) [Table 1]
C.	**Thickness Measurements**	-	Reference of TM Report
		-	Summary of where measured
		-	Separate form indicating the tanks/areas with Substantial Corrosion, and corresponding diminution
D.	**Tank Protection**		Separate form indicating [Table 2] :
		-	Location of coating
		-	Condition of coating
E.	**Conclusion**	-	Statement on evaluation/ verification of survey report

Form-WP-016-1

Naval Architect & Offshore Engineering, FSRU, FSO, FPSO Supervision, Ship Survey & Assessment, Ship Maintenance Management, Ship Re-drawing, Cargo Gear Load Test, Hull Thickness Measurement, Non-Destructive Test, Rope Access Services, Underwater Survey/ROV/salvage and Drone Survey. Approved by: ABS, BKI, BV, DNV-GL, IRS, LR, KR, NK, RINA

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TABLE 1
CLOSE-UP SURVEY

TANK	FRAME NO.	STRUCTURAL MEMBER DESCRIPTION	FINDING	PICTURE

TABLE 2
TANK PROTECTION

TANK [1]	TANK PROTECTION [2]	COATING CONDITION [3]	REMARKS

Remarks:

[1] All segregated ballast tanks and combined cargo/ballast tanks to be listed
[2] C = Coating NP = No Protection
[3] Coating condition according to the following standard

GOOD condition with only minor spot rusting

FAIR condition with local breakdown at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition.

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

Form-WP-016-1